Program 7:
Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle

Step 1: To build a basic Java Application with Maven Project, run following command:
mvn archetype: generate -DgroupId=com.example -DartifactId=MyMaventoGradle -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
shreya@ShreyaM:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactiId=maven-archetype-quickstart -Di
nteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 3.4 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/shreya/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenToGradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/shreya/devops/MyMavenToGradle
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  19.433 s
[INFO] Finished at: 2026-03-25T08:57:31+05:30
[INFO] ------------------------------------------------------------------------
shreya@ShreyaM:~/devops$
```

Step 2: enter the MyMaventoGradle directory and open the pom.xml file.

```
shreya@ShreyaM:~/devops/MyMavenToGradle$ gedit pom.xml
shreya@ShreyaM:~/devops/MyMavenToGradle$
```

Add the code in the pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>

  <artifactId>MyMavenApp</artifactId>

  <packaging>jar</packaging>

  <version>1.0-SNAPSHOT</version>

 <name>MyMavenApp</name>

  <url>http://maven.apache.org</url>
```

```
  <!-- Properties: Customize Java version or plugin versions -->
  <properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
</properties>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
```

```
<version>2.22.2</version>
</plugin>
<plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-jar-plugin</artifactId>
        <version>3.1.0</version>
        <configuration>
            <archive>
                <manifestEntries>
                    <Main-Class>com.example.MainClass</Main-Class>
                </manifestEntries>
            </archive>
        </configuration>
    </plugin>
</plugins>
</build>
</project>
```

Step 3 : Run **mvn clean install** to compile your project

```
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/shreya/devops/MyMavenToGradle/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/shreya/devops/MyMavenToGradle/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenToGradle ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenToGradle ---
[INFO] Building jar: /home/shreya/devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenToGradle ---
[INFO] Installing /home/shreya/devops/MyMavenToGradle/target/MyMavenToGradle-1.0-SNAPSHOT.jar to /home/shreya/.m2/repository/com/example/MyMavenToGradle
/1.0-SNAPSHOT/MyMavenToGradle-1.0-SNAPSHOT.jar
[INFO] Installing /home/shreya/devops/MyMavenToGradle/pom.xml to /home/shreya/.m2/repository/com/example/MyMavenToGradle/1.0-SNAPSHOT/MyMavenToGradle-1.
0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.648 s
[INFO] Finished at: 2026-03-25T09:21:17:05:30
```

Step 4: Type the command **mvn exec:java –Dexec.mainClass=”com.exampls.App”** to execute

```
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:MyMavenApp >-----------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenApp ---
Hello World!
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.662 s
[INFO] Finished at: 2026-03-25T12:10:12+05:30
[INFO] ------------------------------------------------------------------------
```

Step 5: Type the comman **gradle init --type pom.xml, it will give Build Failed .**

```
shreya@ShreyaM:~/devops/MyMavenToGradle$ gradle init --type pom.xml
Starting a Gradle Daemon, 1 incompatible and 1 stopped Daemons could not be reused, use --status for details

FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':init'.
> The requested build setup type 'pom.xml' is not supported. Supported types: 'basic', 'groovy-application', 'groovy-library', 'java-application', 'java
-library', 'pom', 'scala-library'.

* Try:
Run with --stacktrace option to get the stack trace. Run with --info or --debug option to get more log output. Run with --scan to get full insights.

* Get more help at https://help.gradle.org

BUILD FAILED in 13s
2 actionable tasks: 2 executed
shreya@ShreyaM:~/devops/MyMavenToGradle$
```

Step 6: Type the comman **gradle init --type pom** to initialize a Gradle Project using gradle init command with type as pom, command converts pom.xml structure to Gradle’s build.gradle.

```
shreya@ShreyaM:~/devops/MyMavenToGradle$ gradle init --type pom

> Task :init
Maven to Gradle conversion is an incubating feature.


BUILD SUCCESSFUL in 6s
2 actionable tasks: 1 executed, 1 up-to-date
shreya@ShreyaM:~/devops/MyMavenToGradle$ ls
build.gradle  gradle  gradlew  gradlew.bat  pom.xml  settings.gradle  src  target
shreya@ShreyaM:~/devops/MyMavenToGradle$
```

Type the ls command to see all the files and directories.

```
shreya@ShreyaM:~/devops/MyMavenToGradle$ ls
build.gradle  gradlew      pom.xml          src
gradle        gradlew.bat  settings.gradle  target
```

Continuous Integration (CI) Pipeline with Jenkins

MyMavenToGradle in Jenkins:

Step 1: in the terminal type the code MavenToGradle to enter the folder .

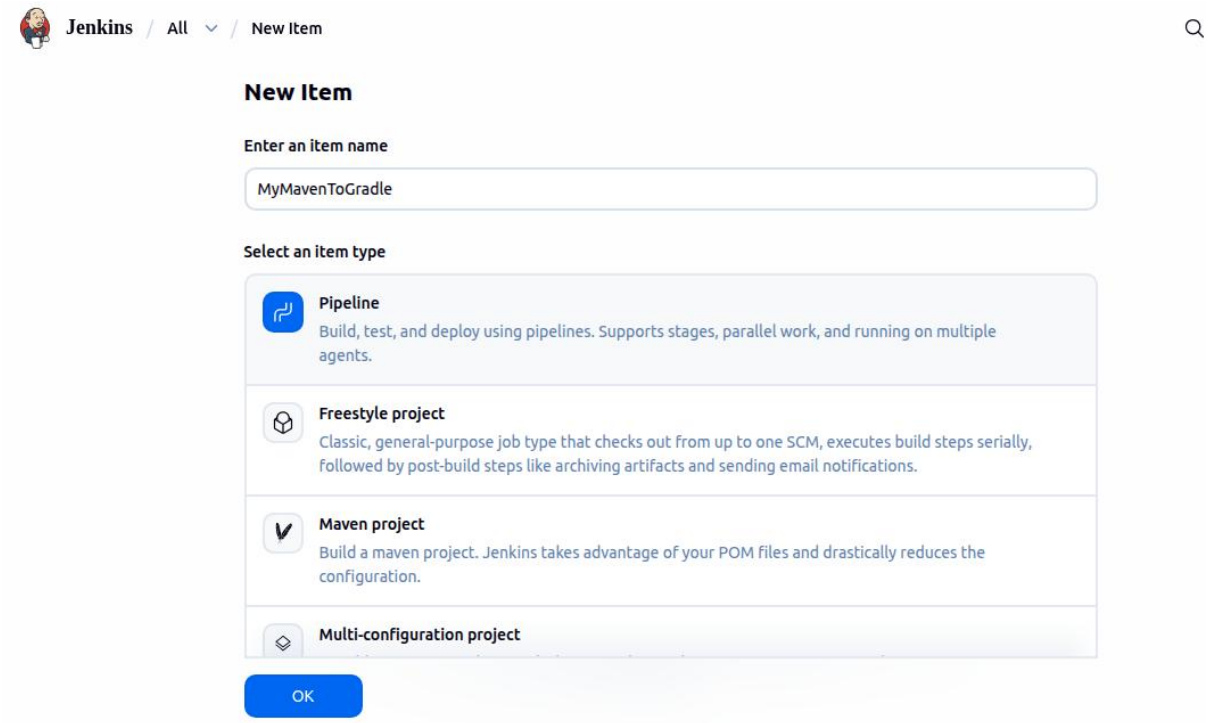
Type the command gedit JenkinsFile to open the Jenkinsfile and paste the code

```
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/shreya20m/maventogradle.git'
            }
        }

        stage('Build') {
            steps {
                sh 'chmod +x gradlew'
                sh './gradlew clean build'
            }
        }

        stage('Test') {
            steps {
                sh './gradlew test'
            }
        }

        stage('Run Application') {
            steps {
                sh './gradlew run'
```

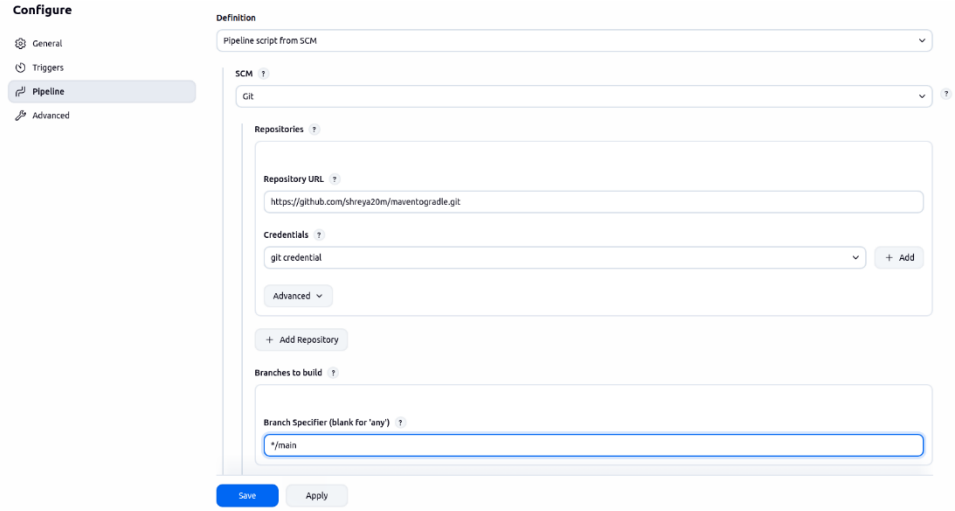
```
            }
        }
    }

    post {
        success {
            echo 'Build successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 2: in Jenkins Dasboard click on New Item

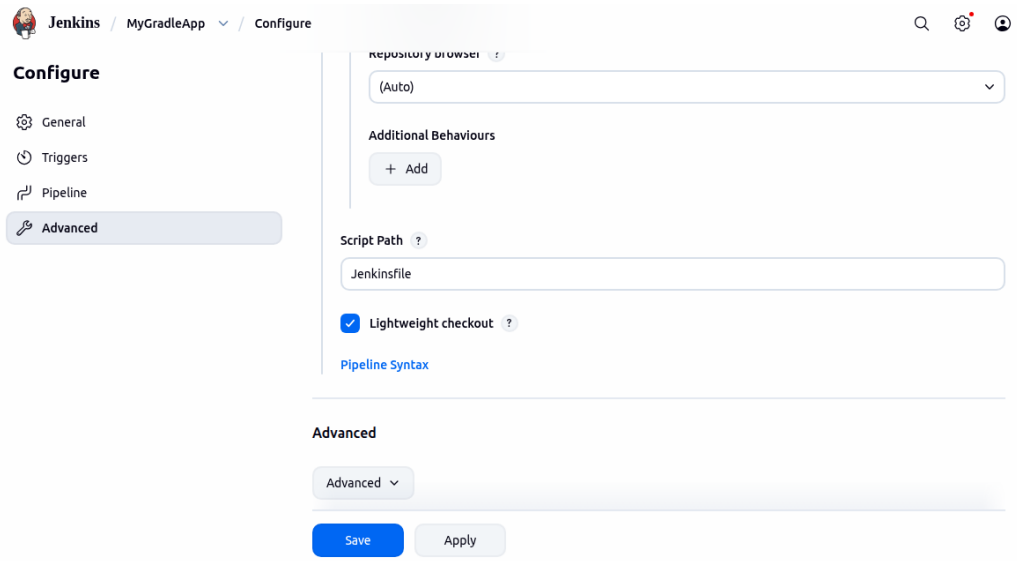
Give the name MyMavenToGradle and select Pipeline



In configure scroll down to Pipeline and choose pipeline script from SCM,give the repository url and choose GitHub Credential



Then click on apply and save



Step 3: go to Jenkins Dasboard select the MyMavenToGradle pipeline and select Build Now. After build select Full Stage View. Once the errors are corrected Build again

Devops

MyMavenToGradle - Stage View

	Declarative: Checkout SCM	Checkout	Build	Test	Run Application	Declarative: Post Actions
Average stage times: (full run time: ~4min 30s)	10s	12s	1min 22s	3s	2s	1s
#4 Apr 14 15:20 — 1 commit	5s	3s	3min 20s	9s	6s	453ms
#3 Apr 14 14:34 — 2 commits	17s	30s	18s failed	553ms failed	546ms failed	1s
#2 Apr 14 14:27 — 1 commit	6s	6s	14s failed	1s failed	1s failed	2s
#1 Apr 14 14:07 — No Changes	10s	8s	1min 34s failed	2s failed	890ms failed	3s